

Data terminology

A

Analyze the data

The analyze the data step is the fourth step in the data analysis process, where data analysts apply statistical and analytical techniques to the cleaned and organized data in order to uncover patterns, trends, and insights.

Artificial intelligence

Artificial intelligence (AI) refers to the ability of a machine to learn patterns and make predictions.

D

Data

Data is plural for datum, which is any piece of information that you can collect, analyze, and interpret to make decisions. Data is a collection of information.

Database

A database is an organized collection of data stored in a computer system.

Data analysis

Data analysis refers to the process of examining raw data to extract meaningful insights that can guide decision making.

Database management system

Database management system (DBMS) refers to software that helps you manage and manipulate databases.

Data overload

Data overload refers to a situation where the volume of data becomes so large and complex that it overwhelms the systems and people responsible for handling it.

G

Gather the data

The gather the data step is the second step in the data analysis process, where data analysts collect the necessary data needed to address the defined objectives.

I**Interpret the results**

The interpret the results step is the final step in the data analysis process, where data analysts make sense of the insights uncovered during the data analysis.

IoT

Internet of Things (IoT) refers to the network of connected devices that collect, share, and exchange data.

M**Manipulate the data**

The manipulate the data step is the third step in the data analysis process, where data analysts transform and prepare the collected data for analysis.

P**Prepare for analysis**

The prepare for analysis step is the first step in the data analysis process, where data analysts focus on defining the objectives, understanding what is needed for the analysis, and creating a plan that ensures the process runs smoothly.

Primary data

Primary data is data that researchers or someone that they directly work with collect for their own purposes, such as answering a business-related question. Researchers usually collect primary data through methods such as surveys, experiments, observations, or interviews.

Q**Qualitative data**

Qualitative data, or categorical data, is descriptive data that represents non-numeric characteristics such as appearances, experiences, opinions, behaviors, attitudes, or beliefs.

Quantitative data

Quantitative data, or numerical data, is directly measurable data that is numerical in nature.

S**Secondary data**

Secondary data is data that another person, department, company, or organization collects for their own purposes, not the researcher's. Researchers use secondary data to help them reach their goals. Frequently used secondary data sources include government reports, online databases, and private businesses.

Structured data

Structured data refers to information that is organized in a clear, defined manner, typically in rows and columns.

SQL

Structured query language (SQL) is a programming language that allows users to ask questions or “query” the database and modify its contents.

U**Unstructured data**

Unstructured data refers to information that does not follow a predefined format and lacks any built-in organization or structure.